PENETRATION TESTING REPORT

Footprinting, Reconnaissance & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083 – Networkwalks
Date	5 October 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (GHDB) W2-PM3 (Maltego) W2-PM4 (theHarvester) W2-PM5 (Zenmap Scanning)
Client / Target	1. networkwalks.com – assigned educational reconnaissance target 2. My own local LAN
Permission obtained?	Yes – all work was carried out inside the assigned educational scope and on my own local network.
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phases 3–5: Reconnaissance exercises and supporting information gathering

1. Liability Disclaimer

All activities in this report were carried out strictly inside the assigned educational cybersecurity scope, and only on systems and devices that I own or have been authorised to assess. The material is provided for learning and research purposes alone and must not be used to break any law. Neither the instructor, the authors nor Networkwalks accept responsibility for any misuse of this knowledge. Anyone who acts outside an authorised scope is personally accountable for those actions, and unauthorised access can be a criminal offence even if no damage results.

2. Introduction

This report documents the practical work I completed in Week 2 of the Cybersecurity & Ethical Hacking program at Networkwalks. It includes footprinting of the networkwalks.com domain with several Kali Linux tools (W2-PM1), Google Hacking Database reconnaissance (W2-PM2), relationship analysis in Maltego (W2-PM3), passive reconnaissance with theHarvester (W2-PM4) and network discovery using Zenmap (W2-PM5).

Everything was done inside Kali Linux. The reconnaissance tasks relied on publicly available information, while the network-scanning task was run against my own local LAN. For each activity, the report states the command or method used, the result observed and a brief note on why that result matters for cybersecurity. Screenshots that support the work are collected in Section 8, each with a short description.

3. Tools Used

The table below summarises each tool used in this report and what it was used for.

Tool	Purpose
Kali Linux	Operating system used for reconnaissance, passive information gathering and network scanning.
WHOIS	Retrieves public domain registration details, key dates and name servers.
WhatWeb	Fingerprints the web technologies, CMS and plugins visible on a site.
nslookup	Resolves a domain name to its IP address through DNS.
curl -I	Fetches the HTTP response headers returned by the website.
wafw00f	Checks whether a Web Application Firewall can be identified publicly.
dnsrecon	Enumerates DNS records such as NS, MX, SPF, TXT and service records.
Google / GHDB	Carries out search-engine reconnaissance using specialised search operators.
Maltego	Builds a visual graph of entities and relationships and runs reconnaissance transforms.
theHarvester	Gathers public emails, hosts, IPs and related data from supported sources.
Zenmap / Nmap	Discovers live hosts on the authorised local subnet and draws the network topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance

Reconnaissance on the networkwalks.com domain was carried out with six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool gathered a different type of publicly observable information.

WHOIS came first, giving public domain registration details and the domain's name servers, which describe how the domain is registered and hosted. WhatWeb was then used to fingerprint the site's technologies; it reported WordPress 7.0.4 and WP Download Manager 3.3.58 along with other exposed details. With Nslookup, the domain was resolved to the IP address 192.232.216.135.

Curl with the -I option was used to read the HTTP response headers, which added detail about the web application and exposed the WordPress REST API endpoint /wp-json/. Wafw00f was run next to check for a Web Application Firewall, and it identified ModSecurity (SpiderLabs).

Lastly, DNSRecon enumerated the DNS records, returning data on name servers, mail servers, SPF/TXT records, service records and the DNS software in use.

4.2 Google Hacking Database (GHDB)

The GHDB activity introduced reconnaissance through search engines using specialised Google operators. The practical focused on finding publicly indexed security-camera results and open directory listings that contain mathematics PDF material. The captured results were recorded in the accompanying worksheet.

For the security-camera search, ten results were recorded from the captured Google output. They came from sources such as Alliance for Coastal Technologies, 100.42.92, Course Hero, Exploit-DB, Accio Work, Blogger and xvideosbr.to. Only the sources of the results are listed; no direct camera-access URLs or credentials are reproduced.

#	Source	Recorded Search Result
1	Alliance for Coastal Technologies	Live view – AXIS 215 PTZ Network Camera
2	100.42.92	Live View – AXIS 211 Network Camera version 4.11
3	Course Hero	Cameras-Long.txt — ViewerFrame / Live View – AXIS
4	Exploit-DB	tilt title: Live View / – AXIS
5	Exploit-DB	intitle: MJPG Live Demo / intext: Select preset position
6	Accio Work	Live View Axis: Find Top PTZ Cameras & Streaming Devices
7	Blogger.com	Inurl Live View Axis Night Club Anyone Want To Hack Into …
8	Exploit-DB	intitle: Live View / – AXIS — Various Online Devices GHDB
9	xvideosbr.to	Site query involving inurl:view.shtml / intitle: Live View / – AXIS
10	Exploit-DB	tilt intitle: Live View / – AXIS | inurl:view/view.shtml

The mathematics directory-listing exercise gave the following ten documented entries.

#	Source	Directory Listing	URL
1	University of New Mexico	Index of ~/megrad/Math	http://www.unm.edu/~megrad/Math/
2	Ochicken.net	Index of /library/Mathematics/	https://ochicken.net/library/Mathematics/
3	University of Liverpool	Index of /maths	http://pcwww.liv.ac.uk/maths/
4	The Netlib	Index of /math/docpdf	https://www.netlib.org/math/docpdf/
5	ИПТ РАН	Index of /ebooks/books/open	http://www.iisp.ac.ru/ebooks/books/open/
6	University of Galway	Index of ~/rquinlan/linearalgebra	https://maths.nuigalway.ie/~rquinlan/linearalgebra/
7	Case Western Reserve University	Index of /artsci/math/singer/publish	https://case.edu/artsci/math/singer/publish/

#	Source	Directory Listing	URL
8	Dartmouth Mathematics	Index of ~/carlp/PDF	https://www.math.dartmouth.edu/~carlp/PDF/
9	Jsoftware	Index of /books/pdf	https://www.jsoftware.com/books/pdf/
10	Carnegie Mellon University	Index of /users/math/wow/osummerinst	https://www.cmu.edu/users/math/wow/osummerinst/

4.3 Maltego

In Maltego, a Domain entity was created for networkwalks.com and the transform [Utilities] To Emails @domain [Search Engine] was run on it. The transform completed on one entity and consumed 10 unified credits, leaving 190. The graph stayed centred on the networkwalks.com entity, and no further linked email entity appeared in the captured graph.

Item	Recorded Result
Domain entity	networkwalks.com
Transform	[Utilities] to emails @domain [Search Engine]
Entities processed	1
Execution status	Completed
Credits consumed	10 unified credits
Credits remaining	190
Visible graph result	Domain node remained visible; no additional linked entity appeared in the captured graph.

4.4 Footprinting with theHarvester

For the theHarvester exercise, microsoft.com was the passive reconnaissance target specified in the module. Task 1 used Baidu as the source with a limit of 1000 results. Task 2 used all supported sources with a limit of 50. The two outputs were saved separately as raw text, since the all-source output was too long to include in full in this report.

Task	Command	Observed Result
Task 1	theHarvester -d microsoft.com -l 1000 -b baidu	0 IPs, 0 emails and 0 people found; 2 hosts found: cla.opensource.microsoft.com and support.microsoft.com.
Task 2	theHarvester -d microsoft.com -l 50 -b all	The all-source passive reconnaissance output was saved separately as raw output. Because it was very large, the full terminal output is not reproduced here.

4.5 Network Scanning with Zenmap

The final activity used Zenmap to perform network discovery on my own local network. The task involved finding my local IP address and subnet, discovering live hosts, noting their IP and MAC addresses, and generating a network topology.

I began with the Kali ip command to find the active interface and its IPv4 address. The eth0 interface held 10.0.0.3/24, which places it in the 10.0.0.0/24 subnet.

I then entered that subnet into Zenmap and chose the Ping Scan profile. Zenmap ran nmap -sn 10.0.0.0/24, scanned 256 IP addresses and found three live hosts.

IP Address	MAC Address	Status
10.0.0.1	52:54:00:12:35:00	Up
10.0.0.2	08:00:27:53:38:90	Up
10.0.0.3	08:00:27:5A:87:BC	Up – Kali host

The MAC address of 10.0.0.3 was also verified directly on the eth0 interface using ip link show eth0. Once the scan had finished, I opened the Topology tab in Zenmap and generated the network topology.

5. Risk Analysis / Impact

From the reconnaissance, GHDB, passive harvesting and local network scanning activities, the following potential risks were identified. They are observations from information gathering and host discovery, not confirmed vulnerabilities.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology details exposed	WhatWeb identified WordPress and WP Download Manager	Attackers can use visible technology and version data to pick software worth targeting	● Medium
2	Server IP address discoverable	Nslookup resolved the domain to 192.232.216.135	Reveals where the web service sits on the network	● Low
3	HTTP technical details exposed	Curl returned response headers and revealed /wp-json/	Can support technology fingerprinting and deeper enumeration	● Low
4	WAF identifiable	Wafw00f detected ModSecurity (SpiderLabs)	Discloses information about the application's defensive set-up	● Low
5	DNS infrastructure details exposed	DNSRecon listed DNS, mail and service-related records	DNS data helps an outsider build a wider infrastructure profile	● Medium
6	Publicly indexed information discoverable	GHDB searches returned camera-related and directory-listing results	Search-engine indexing can expose data that aids profiling and reconnaissance	● Medium

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
7	Public-source host data collected	theHarvester found two microsoft.com hostnames in the Baidu task	Public host data supports external reconnaissance and infrastructure mapping	● Low
8	Several live hosts on local network	Zenmap found three live hosts in 10.0.0.0/24	Unknown or unauthorised devices could be present on a network	● Medium

Risk level key: ● **Critical** ● **Medium** ● **Low**

These risks come from reconnaissance and scanning exercises and should not be read as confirmed vulnerabilities. The practicals mainly involved information gathering, public-source enumeration and host discovery, and no exploitation or vulnerability validation took place.

For that reason, the visibility of a software version, IP address, DNS record, indexed search result or hostname does not on its own mean a system is vulnerable. Any real vulnerability would have to be confirmed through further authorised security testing.

6. Recommendations

Based on what was observed in these activities, I suggest the following security improvements:

1. **Review publicly exposed technology information.** Regularly check which web technologies, CMS platforms and plugins are visible to outsiders.
2. **Keep software updated.** Update CMS platforms, plugins and other components regularly and compare them with current security advisories.
3. **Review HTTP headers.** Examine response headers to make sure no unnecessary technical details are being revealed.
4. **Review DNS records regularly.** Audit DNS records periodically so that only required information and services are public.
5. **Configure and monitor the WAF properly.** Keep the WAF switched on and tuned, and review its rules and logs on a regular basis.
6. **Review search-engine indexing.** Check search results from time to time for content that should not be publicly indexed.
7. **Review public directories and files.** Inspect open directory listings and files, and restrict any that are not meant to be public.
8. **Review external information exposure.** Run passive reconnaissance periodically to see what infrastructure data an outside observer can collect.
9. **Perform regular internal network discovery.** Scan your own networks at intervals to identify which devices are active.
10. **Investigate unknown devices.** Look into any unexpected device found during scanning and confirm whether it is legitimate.

11. **Maintain network documentation.** Keep network topology and device inventories documented and up to date.
12. **Test security only with authorisation.** Carry out reconnaissance and scanning only on systems and networks where proper permission exists.

7. Conclusion

In Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical work on footprinting, reconnaissance, search-engine intelligence gathering, passive harvesting and network scanning.

During footprinting, six Kali Linux tools were used to collect information on networkwalks.com. I learned that WHOIS supplies domain information, WhatWeb identifies web technologies, Nslookup resolves domain names, Curl exposes HTTP headers, Wafw00f detects a WAF and DNSRecon provides further DNS data.

The GHDB activity showed how search-engine operators can find publicly indexed information. I recorded ten security-camera-related results and ten mathematics directory listings, which demonstrated how search engines can unintentionally expose data that helps an attacker's reconnaissance.

In the Maltego exercise, I made a domain entity for networkwalks.com and ran a search-engine email-domain transform. It completed successfully, though no extra linked entity showed up in the captured graph.

The theHarvester exercise illustrated passive reconnaissance with different public-source settings. The Baidu task returned two hostnames and no IP, email or people results, whereas the all-source task produced a much larger output that was kept separately as raw evidence.

Finally, Zenmap helped me work out my local network configuration and find active hosts. The Kali interface was 10.0.0.3/24, the scanned subnet was 10.0.0.0/24 and three live hosts were found. I also collected IP and MAC address details and produced a network topology.

These exercises showed me that information gathering is a core part of cybersecurity: even before trying to exploit anything, a professional can learn a great deal about an environment by carefully analysing public information and network responses.

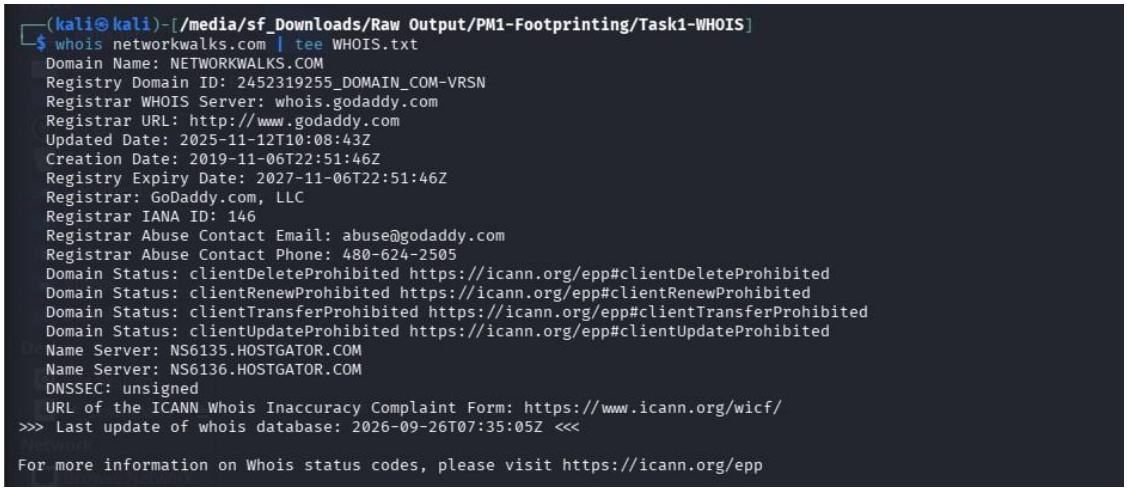
I also learned that findings must be documented clearly. A good report explains what was done, what was found, what it means, what risk it could create and how that risk can be reduced.

Above all, I learned that reconnaissance and scanning must always stay within an authorised scope. All of this work was done as part of the assigned educational lab and on my own local network.

8. Evidences Collected

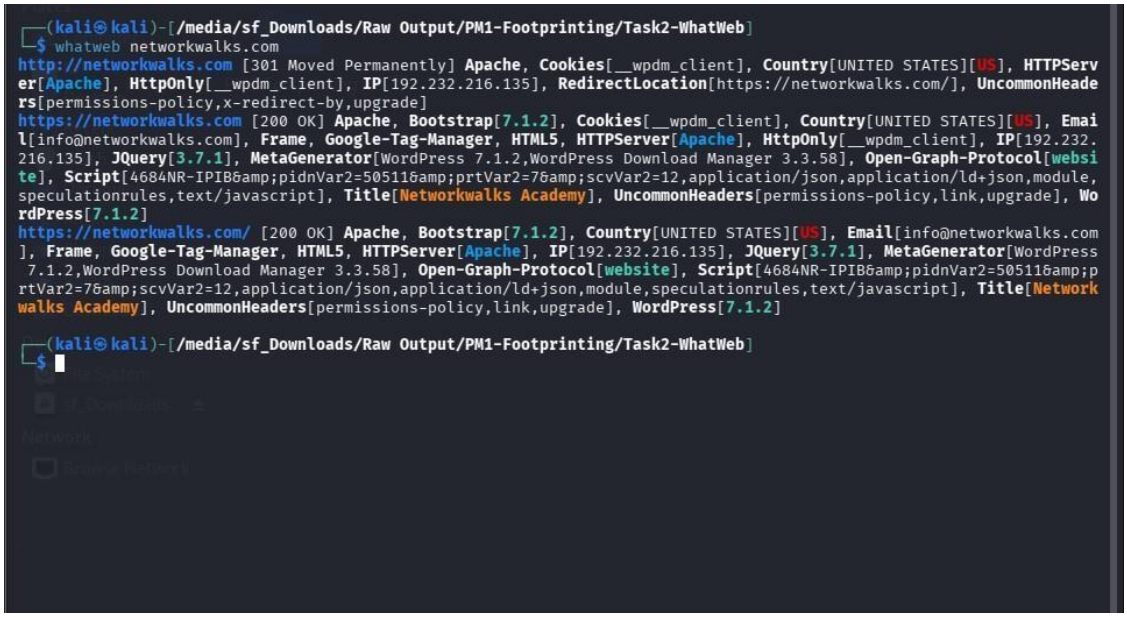
The screenshots below support the activities described in Section 4. Each is followed by a short description of what it shows.

Screenshot 1: PM1 — WHOIS output



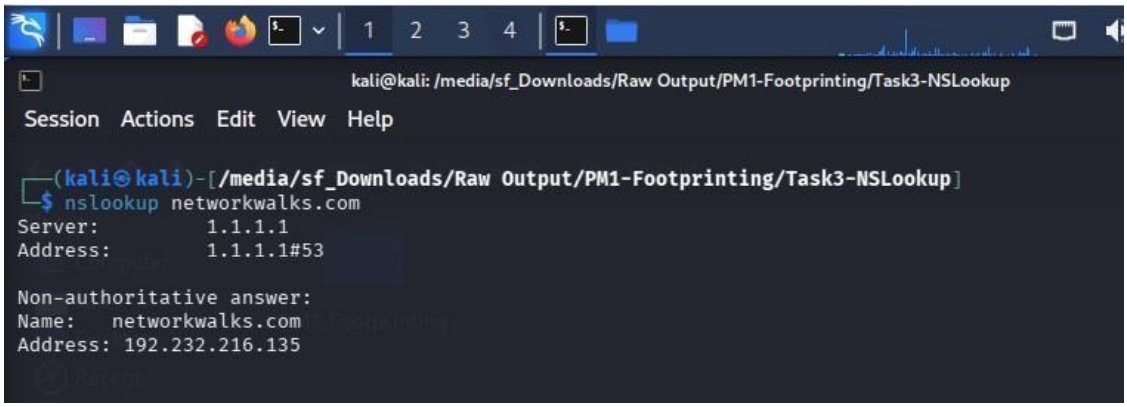
Description: *This screenshot shows the WHOIS lookup for networkwalks.com. It lists the registrar, registration and expiry dates, and the name servers. These details outline who manages the domain and where it is hosted.*

Screenshot 2: PM1 — WhatWeb output



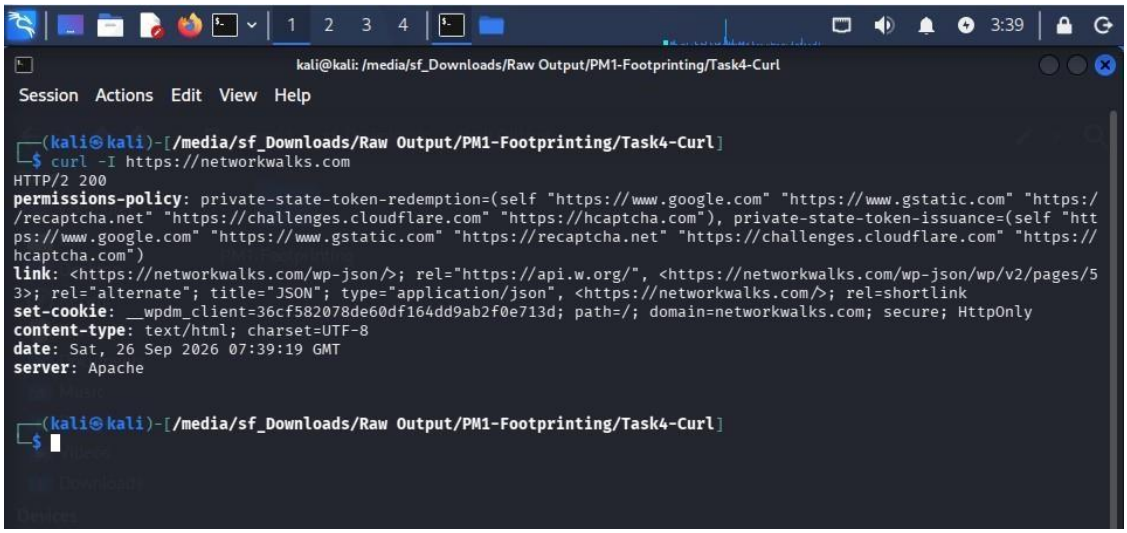
Description: *WhatWeb fingerprinted the site and returned its technology stack, including WordPress and the WP Download Manager plugin. Version information like this is useful for judging which components need security review.*

Screenshot 3: PM1 — Nslookup output



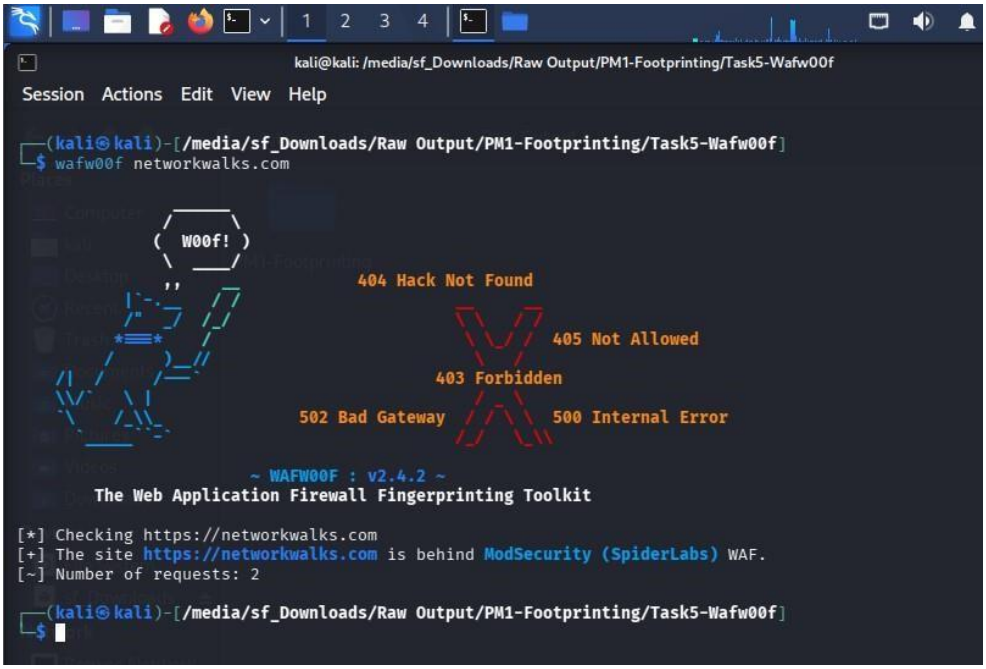
Description: *Nslookup converted the domain name into its IP address, 192.232.216.135. This confirms where the web service is reachable on the network.*

Screenshot 4: PM1 — Curl HTTP headers



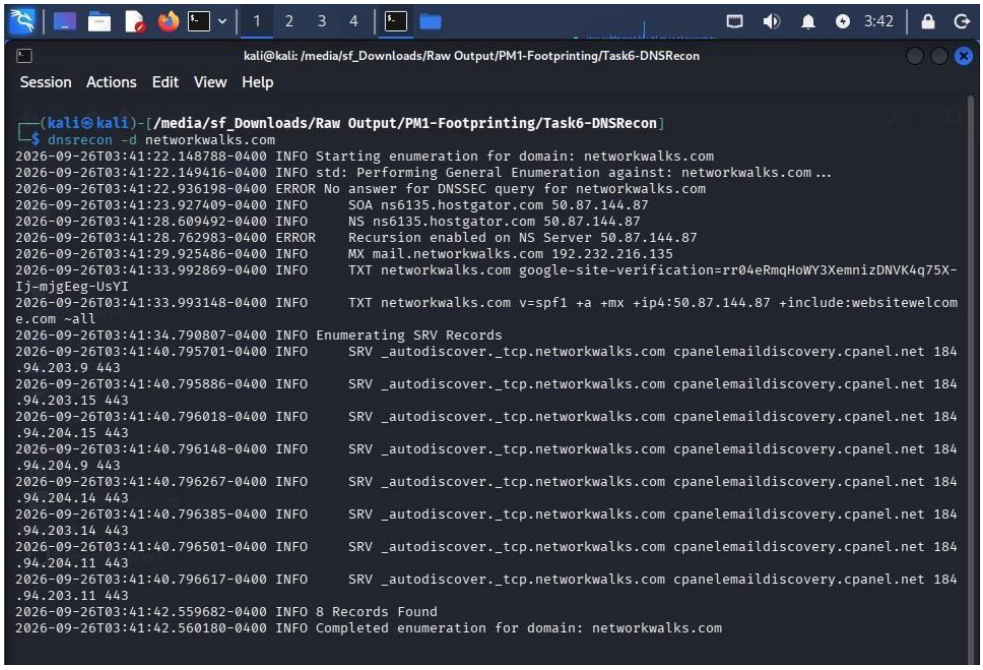
Description: *The curl -I command displayed the HTTP response headers sent by the site. They give extra detail about the web application and show the exposed WordPress REST API path /wp-json/.*

Screenshot 5: PM1 — Wafw00f output



Description: *Wafw00f tested the website for a Web Application Firewall and identified ModSecurity (SpiderLabs). Knowing the WAF in use reveals part of the site's defensive architecture.*

Screenshot 6: PM1 — DNSRecon output



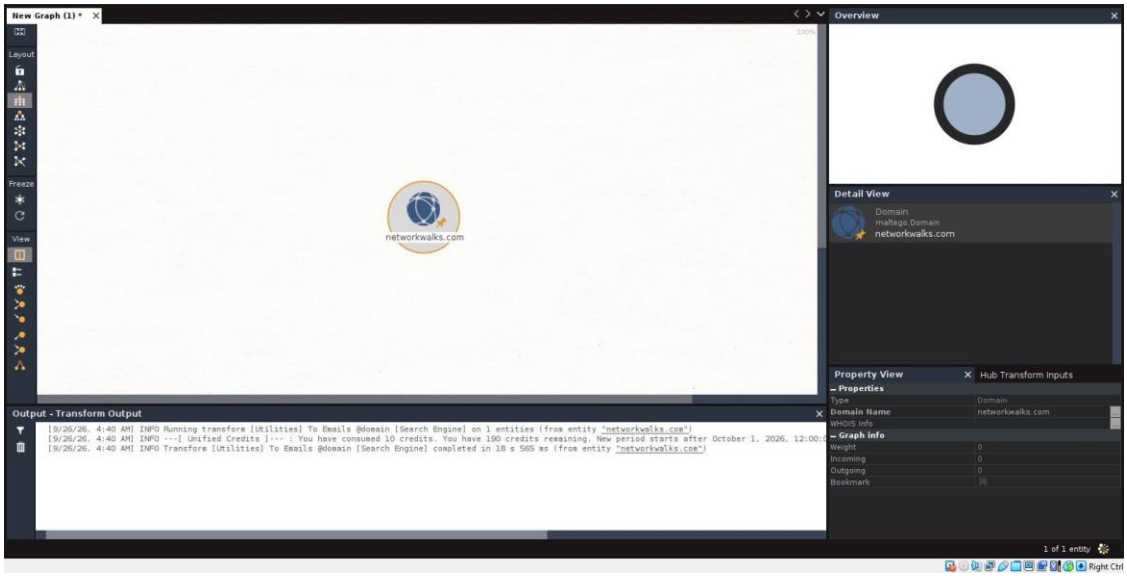
Description: *DNSRecon enumerated the domain's DNS records, covering name servers, mail servers, SPF/TXT entries and service records. Together they provide a wider picture of the DNS infrastructure.*

Screenshot 7: PM3 — Maltego domain graph



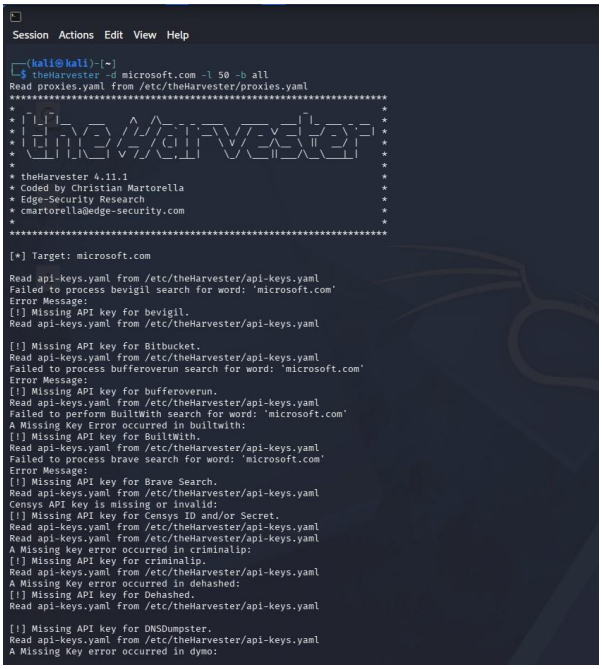
Description: *This is the Maltego graph with networkwalks.com added as a Domain entity. It acts as the starting point for the transform that was run on it.*

Screenshot 8: PM3 — Maltego transform output



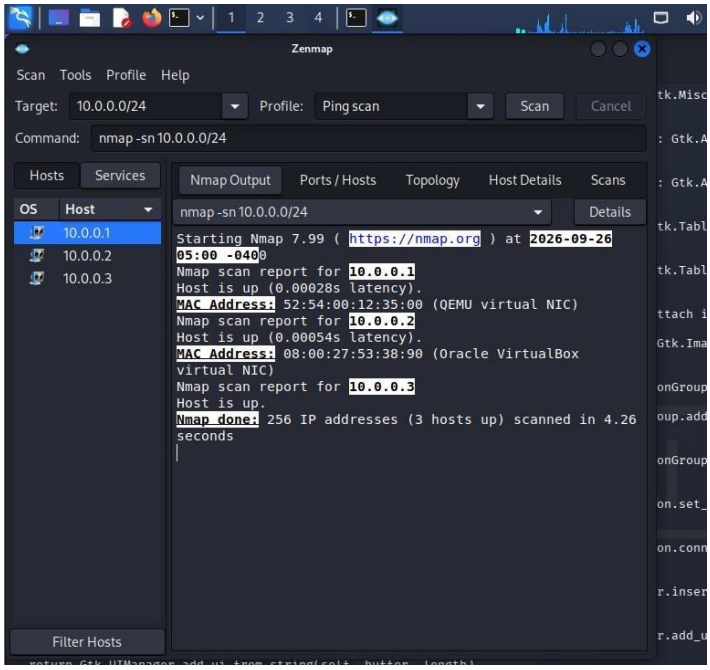
Description: *The output shows the email-domain search-engine transform finishing successfully on one entity. It also records that 10 credits were used and 190 remained, with no extra linked entity appearing.*

Screenshot 9: PM4 — theHarvester all-source raw output



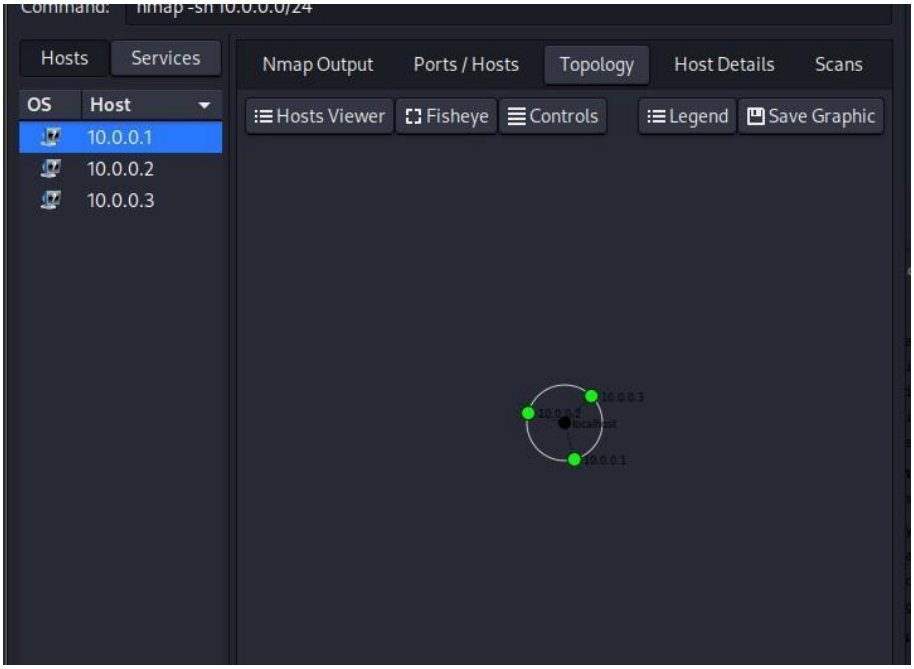
Description: *This is the raw output of theHarvester run against microsoft.com with all sources and a limit of 50. It is lengthy, so it is kept as evidence instead of being copied into the report text.*

Screenshot 10: PM5 — Zenmap Ping Scan output



Description: *Zenmap ran a ping scan (nmap -sn 10.0.0.0/24) across 256 addresses and found three live hosts. The results list each host's IP address and MAC address.*

Screenshot 11: PM5 — Zenmap topology



Description: *The topology view in Zenmap draws the discovered hosts and how they connect on the 10.0.0.0/24 network. It gives a quick visual layout of the local environment.*